

Functions:

```
let rec nrev = function
| [] -> []
| h :: t -> nrev t @ [ h ] (* reverse the tail and append the head *)
```

```
let rec irev acc = function
| [] -> acc
| h :: t -> irev (h :: acc) t
```

```
let rev lst = irev [] lst
```

```
let (@) = append
```

```
let rec append lst1 lst2 =
Match lst1 with
| [] -> lst2
| h :: t -> h :: append t lst2
```

Lemma 1: forall xs ys zs, xs @ (ys @ zs) = (xs @ ys) @ zs

Proof of lemma: By induction on xs.

Base case: xs = []

Show: forall ys zs, [] @ (ys @ zs) = ([] @ ys) @ zs

```
[] @ (ys @ zs)
= { evaluation }
ys @ zs
= { evaluation }
([] @ ys) @ zs
```

Inductive case: xs = h :: t

IH: forall ys zs, t @ (ys @ zs) = (t @ ys) @ zs

Show: forall ys zs, (h :: t) @ (ys @ zs) = ((h :: t) @ ys) @ zs

(h :: t) @ (ys @ zs)	((h :: t) @ ys) @ zs
= { evaluation }	= { evaluation of inner @ }
h :: (t @ (ys @ zs))	(h :: (t @ ys)) @ zs
= { IH }	= { evaluation of outer @ }
h :: ((t @ ys) @ zs)	h :: ((t @ ys) @ zs)

QED

Lemma 2: forall h, forall t, [h] @ t = h :: t

Proof of lemma: By case analysis on t

Case 1: $t = []$

Show: $\text{forall } h, [h] @ [] = h :: []$

$[h] @ []$
= { evaluation }
 $h :: [] @ []$
= { evaluation of @ }
 $h :: []$
= { evaluation }
 $[h]$

$h :: []$
= { evaluation }
 $[h]$

Case 2: $t = \text{hd} :: \text{tl}$

Show: $\text{forall } h, [h] @ (\text{hd} :: \text{tl}) = h :: (\text{hd} :: \text{tl})$

$[h] @ (\text{hd} :: \text{tl})$
= { evaluation of @ }
 $h :: ([] @ (\text{hd} :: \text{tl}))$
= { evaluation of @ }
 $h :: (\text{hd} :: \text{tl})$

QED

Lemma 3: $\text{forall } \text{lst}$ (here formatted as $h :: t$), where lst can be $[]$, $\text{irev } h \ t = \text{nrev } t @ h$

Proof of lemma: By induction on lst .

$P(\text{lst}) = \text{irev } h \ t = \text{nrev } t @ h$

Base case: $\text{lst} = []$

Show: $\text{irev } [] \ [] = \text{nrev } [] @ []$

$\text{irev } [] \ []$
= { evaluation }
 $[]$

$\text{nrev } [] @ []$
= { evaluation of nrev }
 $[] @ []$
= { evaluation of @ }
 $[]$

Inductive case: $\text{lst} = h :: t$

IH: $\text{forall } t = \text{hd} :: \text{tl}, \text{irev } \text{hd } \text{tl} = \text{nrev } \text{tl} @ \text{hd}$

Show: $\text{forall } t = \text{hd} :: \text{tl}, \text{irev } h \ t = \text{nrev } t @ [h]$

$\text{irev } [h] \ t$
= { evaluation }
 $\text{irev } (\text{hd} :: h) \ \text{tl}$
= { IH with $h := \text{hd} :: h$ }
 $\text{nrev } \text{tl} @ (\text{hd} :: h)$

$\text{nrev } t @ [h]$
= { evaluation of nrev }
 $(\text{nrev } \text{tl} @ [\text{hd}]) @ [h]$
= { lemma 1 }
 $\text{nrev } \text{tl} @ ([\text{hd}] @ [h])$
= { lemma 2 }

$\text{nrev tl @ (hd :: h)}$

Theorem: forall lst, $\text{rev lst} = \text{nrev lst}$

Proof of lemma: By case analysis on lst.

$P(\text{lst}) = \text{rev lst} = \text{nrev lst}$

Case 1: $\text{lst} = []$

Show: $\text{rev } [] = \text{nrev } []$

$\text{rev } []$
= { evaluation }
 $\text{irev } [] []$
= { evaluation }
 $[]$

$\text{nrev } []$
= { evaluation }
 $[]$

Case 2: $\text{lst} = h :: t$

Show: $\text{rev } h :: t = \text{nrev } h :: t$

$\text{rev } h :: t$
= { evaluation }
 $\text{irev } [] (h :: t)$
= { evaluation }
 $\text{irev } (h :: []) t$
= { evaluation }
 $\text{irev } [h] t$
= { evaluation }
 $\text{irev } (hd :: h) tl$

$\text{nrev } h :: t$
= { evaluation }
 $\text{nrev } t @ [h]$
= { evaluation }
 $\text{nrev } (tl @ [hd]) @ [h]$
= { lemma 1 }
 $\text{nrev } tl @ [(hd) @ [h]]$
= { lemma 2 }
 $\text{nrev } tl @ (hd :: h)$
= { lemma 3 }
 $\text{irev } (hd :: h) tl$

QED